



TASK 2 REPORT
NETWORK SECURITY AND SCANNING

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1. INTRODUCTION

Network security is an important area of cybersecurity that focuses on protecting computer systems, networks, devices and data from unauthorized access and attacks.

As part of Task 2 of the ApexPlanet Software Cybersecurity Internship, reconnaissance, network scanning, vulnerability assessment, packet analysis and basic firewall configuration were studied and practiced in a controlled virtual laboratory environment.

The task focused on understanding how security professionals collect information about systems, identify open ports and services, detect vulnerabilities, analyze network traffic and apply basic security controls.

All security testing was performed only against controlled laboratory systems.

2. OBJECTIVES

The main objectives of Task 2 were:

- ❖ To understand passive and active reconnaissance.
- ❖ To learn information-gathering techniques such as WHOIS and NSLookup.
- ❖ To understand Google Dorking and Shodan.
- ❖ To perform active reconnaissance using Ping Sweep and Banner Grabbing.
- ❖ To perform TCP and UDP scans using Nmap.
- ❖ To detect service versions using Nmap.
- ❖ To identify operating systems using Nmap.
- ❖ To create a scan report for target systems.
- ❖ To set up a vulnerability scanner such as OpenVAS or Nessus Essentials.
- ❖ To scan a vulnerable test machine.
- ❖ To analyze vulnerability reports based on severity.
- ❖ To capture and analyze HTTP, FTP and DNS traffic.
- ❖ To understand how credentials can be exposed through unencrypted FTP traffic.
- ❖ To understand SYN flood attacks in a controlled environment.
- ❖ To analyze packets using Wireshark.
- ❖ To configure basic iptables firewall rules.
- ❖ To understand how firewall rules can block unwanted network traffic.
- ❖ To document the findings in a security report.

3. LAB ENVIRONMENT

The practical activities were performed using a virtual cybersecurity laboratory.

Attacker / Security Testing Machine

Operating System: Kali Linux

Kali Linux was used because it contains various tools required for reconnaissance, network scanning, vulnerability assessment and packet analysis.

Target System

A vulnerable test machine was used for controlled security testing.

Target IP used during network scanning: 192.168.100.20

Main Tools Used

- ❖ WHOIS
- ❖ NSLookup
- ❖ Google Dorking
- ❖ Shodan
- ❖ Ping
- ❖ Nmap
- ❖ Netcat
- ❖ OpenVAS / Greenbone Vulnerability Scanner
- ❖ Wireshark
- ❖ hping3
- ❖ iptables

4. RECONNAISSANCE

Reconnaissance is the first stage of a security assessment.

The purpose of reconnaissance is to collect information about a target before performing scanning or vulnerability testing.

Reconnaissance can generally be divided into:

1. Passive Reconnaissance
2. Active Reconnaissance

5. PASSIVE RECONNAISSANCE

Passive reconnaissance collects information without directly interacting with the target system in a way that generates significant traffic to the target.

The main techniques covered were:

- ❖ WHOIS
- ❖ NSLookup
- ❖ Google Dorking
- ❖ Shodan

5.1 WHOIS

WHOIS is a query service used to obtain registration-related information about Internet domains and IP address allocations.

Information that may be obtained includes:

- ❖ Domain registration information
- ❖ Registrar information
- ❖ Name servers
- ❖ Registration dates
- ❖ Domain status
- ❖ Organization information

WHOIS can help security professionals understand the ownership and infrastructure associated with a domain.

Example:

whois example.com

WHOIS information can be useful during the information-gathering phase of penetration testing.

5.2 NSLOOKUP

NSLookup is a command-line tool used to query DNS records.

It can be used to identify information such as:

- ❖ IP addresses
- ❖ Name servers
- ❖ Mail servers
- ❖ DNS records

Example:

nslookup example.com

DNS reconnaissance helps security professionals understand how a domain's DNS infrastructure is configured.

5.3 Google Dorking

Google Dorking refers to using advanced search operators to locate publicly indexed information.

Common operators include:

site:

filetype:

intitle:

inurl:

For example:

site:example.com filetype:pdf

Google Dorking can help identify publicly accessible documents or pages that were unintentionally indexed.

It should only be used for authorized security assessments.

5.4 Shodan

Shodan is a search engine that indexes information about Internet-connected devices and services.

It can provide information such as:

- ❖ IP addresses
- ❖ Open ports
- ❖ Services
- ❖ Service banners
- ❖ Device information
- ❖ Software versions

Security professionals can use Shodan to understand externally exposed infrastructure.

However, information discovered through Shodan should be treated responsibly and should not be used to access systems without authorization.

6. ACTIVE RECONNAISSANCE

Active reconnaissance involves directly interacting with the target system to gather information.

The main techniques covered were:

- ❖ Ping Sweep
- ❖ Banner Grabbing

6.1 Ping Sweep

A Ping Sweep is used to identify active hosts within a network range.

For example:

```
nmap -sn 192.168.100.0/24
```

The command performs host discovery without conducting a full port scan.

The purpose of a Ping Sweep is to identify which systems are active before performing detailed scanning.

6.2 Banner Grabbing

Banner grabbing is a technique used to identify information about services running on a target system.

It can reveal:

- ❖ Service name
- ❖ Service version
- ❖ Server type
- ❖ Operating system information
- ❖ Software information

Netcat can be used for basic banner grabbing:

```
nc -nv 192.168.100.20 22
```

Banner information can help identify outdated or vulnerable software.

7. NMAP NETWORK SCANNING

Nmap is one of the most commonly used network scanning tools in cybersecurity.

It can be used for:

- ❖ Host discovery
- ❖ Port scanning
- ❖ Service detection
- ❖ Version detection
- ❖ OS detection
- ❖ Network mapping

8. TCP SYN SCAN

A TCP SYN scan was studied using the Nmap -sS option.

```
sudo nmap -sS 192.168.100.20
```

The SYN scan is commonly called a half-open scan.

Instead of completing a full TCP connection, Nmap analyzes the response to the SYN packet.

A typical response can indicate:

- ❖ SYN/ACK → Port is likely open
- ❖ RST → Port is likely closed
- ❖ No response / filtered response → Port may be filtered

The SYN scan is commonly used during network reconnaissance because it can efficiently identify open TCP ports.

9. UDP SCAN

UDP scanning was performed using the -sU option.

```
sudo nmap -sU 192.168.100.20
```

UDP does not use the same connection mechanism as TCP, making UDP scanning different from TCP scanning.

UDP services can include:

- ❖ DNS
- ❖ DHCP
- ❖ SNMP
- ❖ TFTP

UDP scanning is important because vulnerabilities can exist in UDP services even when TCP ports appear secure.

10. SERVICE VERSION DETECTION

Nmap's -sV option can be used to identify the versions of services running on open ports.

```
sudo nmap -sV 192.168.100.20
```

Service version detection can provide information such as:

- Service name
- Software version
- Product information

This information can then be compared with known vulnerabilities.

For example, if an outdated service version is detected, security professionals can check whether the version has known CVEs.

11. OPERATING SYSTEM DETECTION

Nmap can attempt to identify the operating system of a target using the -O option.

```
sudo nmap -O 192.168.100.20
```

OS detection analyzes characteristics of network responses and compares them with known operating-system fingerprints.

This can help determine whether the target is likely running:

- ❖ Linux
- ❖ Windows
- ❖ BSD
- ❖ Other operating systems

OS detection is useful during security assessments because vulnerabilities and security configurations often depend on the operating system.

12. COMBINED NMAP SCAN

Multiple Nmap techniques can also be combined.

Example:

```
sudo nmap -sS -sV -O 192.168.100.20
```

This performs:

- ❖ TCP SYN scanning
- ❖ Service/version detection
- ❖ OS detection

The results can be used to create a network scan report.

13. NMAP SCAN FINDINGS

The controlled target system was found to be active.

The scan identified multiple exposed services.

Some of the identified services included:

Service	Common Port
FTP	21
SSH	22
Telnet	23
SMTP	25

DNS	53
HTTP	80
SMB/NetBIOS	139/445
RPC-related services	Various

The scan also showed that a large number of ports were closed, while several services were accessible.

The presence of multiple exposed services increases the attack surface of a system.

14. NMAP SCAN REPORT

A scan report should contain the following information:

Target

192.168.100.20

Host Status

Host is up

Approximate Latency

0.014 seconds

Port Status

The scan identified multiple open services and approximately 977 closed TCP ports.

Important Exposed Services

- ❖ FTP
- ❖ SSH
- ❖ Telnet
- ❖ SMTP
- ❖ DNS
- ❖ HTTP
- ❖ SMB/NetBIOS
- ❖ RPC-related services

These services should be reviewed to determine whether they are necessary and securely configured.

15. OPENVAS / GREENBONE VULNERABILITY SCANNER

OpenVAS, now commonly associated with the Greenbone Vulnerability Management platform, is a vulnerability assessment tool.

It can identify vulnerabilities in:

- ❖ Operating systems
- ❖ Network services
- ❖ Applications
- ❖ Software packages
- ❖ Network configurations

The scanner works by comparing detected software and configurations against known vulnerability information.

16. OPENVAS SETUP

OpenVAS was configured in the Kali Linux laboratory environment.

The OpenVAS interface was accessed through the browser.

The vulnerability scanner provides functionality for:

- ❖ Target configuration
- ❖ Scan configuration
- ❖ Vulnerability assessment
- ❖ Report generation
- ❖ Severity classification

The scanner was used as part of the vulnerability-assessment stage of the task.

17. VULNERABILITY SCANNING

The vulnerable test system was configured as the target for the vulnerability assessment.

The general process was:

1. Start the vulnerability scanner.
2. Configure the target IP address.
3. Create a scan task.
4. Start the scan.
5. Wait for the vulnerability assessment to complete.
6. Open the generated report.
7. Analyze vulnerabilities based on severity.

The generated report can contain information about vulnerabilities, affected services, severity levels and recommended remediation.

18. VULNERABILITY SEVERITY

Vulnerability scanners generally classify findings according to severity.

Critical

Critical vulnerabilities represent extremely serious security weaknesses that may allow an attacker to gain significant control or cause severe damage.

Immediate remediation is recommended.

High

High-severity vulnerabilities can significantly affect system security.

They may allow unauthorized access, privilege escalation or sensitive information disclosure.

These findings should be addressed quickly.

Medium

Medium-severity vulnerabilities represent security weaknesses that may require specific conditions to exploit.

They should be addressed as part of normal vulnerability management.

Low

Low-severity findings generally have limited security impact but should still be reviewed.

Some low-risk findings can provide useful information to an attacker when combined with other vulnerabilities.

19. ANALYSIS OF VULNERABILITY REPORTS

A vulnerability report should not be evaluated only by the number of findings.

Each vulnerability should be analyzed based on:

- ❖ Severity
- ❖ Affected service
- ❖ Affected software
- ❖ Vulnerability description
- ❖ Possible impact
- ❖ Exploitability
- ❖ Recommended remediation

For example, an exposed outdated service may receive a high severity rating because an attacker could potentially exploit a known vulnerability.

Security teams should prioritize Critical and High findings first, followed by Medium and Low findings.

20. PACKET ANALYSIS WITH WIRESHARK

Wireshark is a network protocol analyzer used to capture and inspect network packets.

It can be used to analyze:

- ❖ TCP
- ❖ UDP
- ❖ HTTP
- ❖ FTP
- ❖ DNS
- ❖ ARP
- ❖ ICMP
- ❖ Other protocols

Packet analysis allows security professionals to understand how systems communicate across a network.

21. HTTP TRAFFIC ANALYSIS

HTTP traffic was studied using Wireshark.

A display filter can be used to show HTTP packets:

http

HTTP traffic can contain:

- ❖ HTTP requests
- ❖ HTTP responses
- ❖ URLs
- ❖ Headers
- ❖ Cookies
- ❖ Form information

Because traditional HTTP does not encrypt traffic, sensitive information may be visible to someone who can capture the traffic.

HTTPS should therefore be preferred for sensitive web communication.

22. FTP TRAFFIC ANALYSIS

FTP traffic was analyzed to understand the security risks of unencrypted file-transfer protocols.

A Wireshark filter can be used:

ftp

FTP traditionally transmits authentication information without encryption.

If FTP credentials are sent in plaintext, packet capture can potentially reveal:

- ❖ Username
- ❖ Password
- ❖ Commands
- ❖ File-transfer information

This demonstrates why secure alternatives such as SFTP or FTPS should be preferred.

23. FILTERING FTP CREDENTIALS

Wireshark can be used to inspect FTP authentication traffic in a controlled laboratory environment.

A filter such as:

```
ftp.request.command == "USER" || ftp.request.command == "PASS"
```

can help identify FTP username and password commands where supported by the capture.

This demonstrates the security weakness of transmitting credentials without encryption.

The technique was performed only for educational purposes in the controlled lab environment.

24. DNS TRAFFIC ANALYSIS

DNS traffic was also studied using Wireshark.

A basic DNS display filter is:

dns

DNS packets can reveal information such as:

- ❖ Query name
- ❖ Query type
- ❖ Response
- ❖ Source IP
- ❖ Destination IP

DNS analysis can help identify unusual queries, suspicious domains and potential malicious activity.

25. SYN FLOOD ATTACK

A SYN flood is a type of Denial-of-Service attack.

It abuses the TCP connection process by sending a large number of SYN requests without completing the connection.

The basic TCP connection process is:

Client → SYN

Server → SYN/ACK

Client → ACK

During a SYN flood, an attacker sends many SYN packets and does not complete the handshake.

The server may maintain many incomplete connections, potentially consuming system resources.

26. SYN FLOOD SIMULATION

The task included studying SYN flood behavior using hping3 in a controlled environment.

The purpose of the simulation is to understand:

- TCP SYN packets
- Half-open connections
- Network traffic patterns
- Effects on a target
- Detection through packet analysis

Any such testing must remain restricted to an isolated laboratory environment and an authorized target.

27. WIRESHARK ANALYSIS OF SYN FLOOD

Wireshark can be used to identify SYN traffic.

A useful filter is:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

This displays TCP SYN packets that have not yet received an ACK.

During a SYN flood simulation, a large number of SYN packets from the same or multiple sources may indicate abnormal traffic.

Security monitoring systems can use such patterns to identify possible denial-of-service activity.

28. FIREWALL BASICS

A firewall controls network traffic based on predefined rules.

Linux systems can use iptables to define rules that allow or deny traffic.

Basic firewall rules can be used to:

- ❖ Allow required ports
- ❖ Block unnecessary ports
- ❖ Restrict access to services
- ❖ Reduce attack surface

29. IPTABLES ALLOW/DENY RULES

An example of allowing SSH traffic is:

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

An example of blocking HTTP traffic is:

```
sudo iptables -A INPUT -p tcp --dport 80 -j DROP
```

To view existing rules:

```
sudo iptables -L -n -v
```

These rules demonstrate the basic principle of controlling network traffic using a firewall.

Important: Firewall rules should be applied carefully because incorrect rules can block legitimate access to the system.

30. BLOCKING A PORT SCAN

A firewall can be used to reduce exposure to unwanted scanning activity.

For example, if a service is not required, access to its port can be blocked.

Example:

```
sudo iptables -A INPUT -p tcp --dport 23 -j DROP
```

This blocks incoming TCP traffic to port 23, which is commonly associated with Telnet.

After applying the rule, a scan can be performed from the authorized testing machine to verify that the service is no longer accessible.

This demonstrates how firewall rules can reduce the attack surface.

31. SECURITY FINDINGS

The assessment highlighted several important network-security risks.

Finding	Security Concern	Recommended Action
Multiple open ports	Increased attack surface	Disable unnecessary services
FTP	Credentials/data may be unencrypted	Use SFTP/FTPS
Telnet	Unencrypted remote access	Replace with SSH
SMB exposure	Potential file-sharing attacks	Restrict access and patch
HTTP	Unencrypted web communication	Use HTTPS
Outdated services	Known vulnerabilities may exist	Patch/update software
Unrestricted network access	Increased exposure	Apply firewall rules
Weak network monitoring	Attacks may go unnoticed	Use monitoring and logging

32. RECOMMENDATIONS

Based on the assessment, the following security practices are recommended:

1. Disable unnecessary services

Only required services should remain active.

2. Replace insecure protocols

Telnet and unsecured FTP should be replaced with secure alternatives.

3. Apply regular patches

Operating systems and applications should be regularly updated.

4. Configure firewalls

Only required ports should be accessible.

5. Use encrypted communication

HTTPS, SSH, SFTP and other secure protocols should be used instead of plaintext alternatives.

6. Monitor network traffic

Wireshark and other monitoring solutions can help detect suspicious activity.

7. Perform regular vulnerability scans

Vulnerability scanners should be used regularly to identify newly discovered weaknesses.

8. Prioritize critical vulnerabilities

Critical and High vulnerabilities should receive immediate attention.

33. ETHICAL CONSIDERATIONS

All reconnaissance, scanning and vulnerability-assessment activities must be performed only against systems for which permission has been obtained.

The techniques discussed in this task can be used by attackers as well as security professionals. Therefore, responsible and authorized use is essential.

The laboratory environment provides a safe way to learn these techniques without affecting real organizations or unauthorized systems.

34. CONCLUSION

Task 2 provided practical knowledge of network security, reconnaissance, scanning and traffic analysis.

Passive reconnaissance techniques such as WHOIS, NSLookup, Google Dorking and Shodan demonstrated how publicly available information can reveal details about systems and infrastructure.

Active reconnaissance techniques such as Ping Sweep and Banner Grabbing demonstrated how direct interaction can provide additional information about network hosts and services.

Nmap was used to understand TCP and UDP scanning, service-version detection and operating-system detection. The scan of the controlled target identified several network services, demonstrating how exposed services increase the attack surface.

OpenVAS provided an introduction to automated vulnerability assessment and vulnerability severity classification. Wireshark demonstrated how network packets can be captured and analyzed, including HTTP, FTP and DNS traffic.

The study of FTP demonstrated the risks of transmitting credentials without encryption, while the SYN flood simulation provided an understanding of TCP-based denial-of-service attacks.

Finally, iptables demonstrated how firewall rules can be used to control network traffic and reduce exposure.

Overall, Task 2 provided practical experience in the fundamental stages of a network security assessment, from reconnaissance and scanning to vulnerability analysis, packet inspection and basic defensive controls.